Practical 1.2: Linux Server Navigation

File system, monitoring, networking and CRD API access

Day 1 · 3 hours · Resilience Academy SDI Training

Reference: `crd.resilienceacademy.ac.tz` | Data: Tanga & Mwanza, Tanzania

Learning Objectives:

- Navigate the Linux file system confidently
- Monitor system resources (CPU, disk, memory)
- Use networking tools to test connectivity
- Query the live CRD API from the command line

File System Navigation

```
# Where am I?
pwd

# List files (long format, show hidden)
ls -la
ls -lh /var/log

# Change directory
cd /tmp
cd ~ # go to home directory
cd - # go back to previous directory
cd .. # go up one level

# Create and manage files
mkdir ~/sdi-training
cd ~/sdi-training
touch notes.txt
echo "Resilience Academy CRD Training" > notes.txt
cat notes.txt

# Copy, move, delete
cp notes.txt notes_backup.txt
mv notes.txt lecture_notes.txt
rm notes_backup.txt

# View file contents
less lecture_notes.txt # scroll with arrows, q to quit
head -5 /etc/passwd # first 5 lines
tail -5 /etc/passwd # last 5 lines
```

System Monitoring

```
# Running processes (press q to quit)
top
# Or the more readable version:
htop # (install with: sudo apt install -y htop)
```

```
# Disk space usage
df -h # all filesystems
du -sh ~/sdi-training # directory size

# Memory usage
free -h

# System information
uname -a # kernel and OS info
hostname # machine name
uptime # how long running

# List running services
systemctl list-units --type=service --state=running
```

Package Management

```
# Update package list
sudo apt update

# Install a package
sudo apt install -y tree

# Show what tree does
tree ~/sdi-training

# Remove a package
sudo apt remove tree

# Check if a package is installed
dpkg -l | grep nginx
```

Networking Commands

```
# Test connectivity
ping -c 4 8.8.8.8
ping -c 4 crd.resilienceacademy.ac.tz

# HTTP requests with curl
curl https://crd.resilienceacademy.ac.tz
curl -I https://crd.resilienceacademy.ac.tz # headers only

# Download a file
wget https://crd.resilienceacademy.ac.tz/favicon.ico

# Show network interfaces and IPs
ip addr show
ip route show

# Check open ports on this machine
ss -tlnp
```

Exercise: Query the CRD API

```
# List all Mwanza datasets on the live CRD
curl -s "https://crd.resilienceacademy.ac.tz/api/v2/datasets/?search=mwanza" \
```

```
  | python3 -m json.tool

# List all Tanga datasets
curl -s "https://crd.resilienceacademy.ac.tz/api/v2/datasets/?search=tanga" \
  | python3 -m json.tool | grep '"title"'

# Download Mwanza businesses as GeoJSON
curl -s "https://crd.resilienceacademy.ac.tz/geoserver/ows?\
service=WFS&version=2.0.0&request=GetFeature\
&typeName=geonode:mwanza_businesses&outputFormat=application/json" \
  -o mwanza_businesses.geojson

# How many features were downloaded?
python3 -c "
import json
with open('mwanza_businesses.geojson') as f:
    data = json.load(f)
print(f'Downloaded {len(data[\"features\"])} Mwanza business features')
"
```

Checkpoint:

- ✓ You can navigate the file system and create/move/delete files
- ✓ `df -h` and `free -h` show readable system info
- ✓ `ping crd.resilienceacademy.ac.tz` succeeds
- ✓ CRD API returns Mwanza and Tanga datasets
- ✓ `mwanza_businesses.geojson` downloaded with 100+ features